

PROGRAMMING FUNDAMENTALS

Spring 2024

14th March

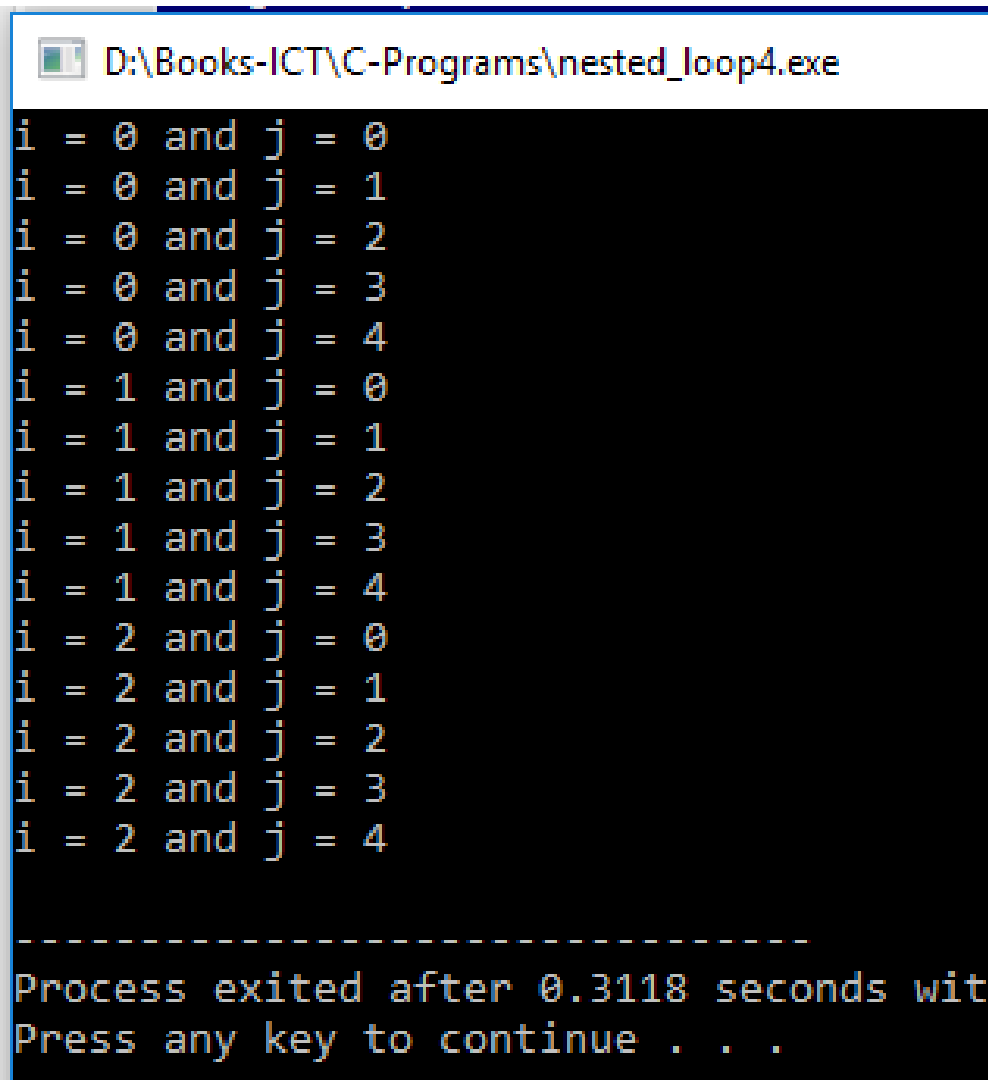
NESTED LOOPS.. Let's practice dry run
A loop within a loop

Let's dry run

```
#include <iostream>
using namespace std;
int main ()
{
    int i = 0;

    while(i < 3)
    {
        int j = 0;
        while(j < 5)
        {
            cout << "i = " << i << " and j = " << j << endl;
            j++;
        }
        i++;
    }
    return 0;
}
```

Running it ...



```
D:\Books-ICT\C-Programs\nested_loop4.exe
i = 0 and j = 0
i = 0 and j = 1
i = 0 and j = 2
i = 0 and j = 3
i = 0 and j = 4
i = 1 and j = 0
i = 1 and j = 1
i = 1 and j = 2
i = 1 and j = 3
i = 1 and j = 4
i = 2 and j = 0
i = 2 and j = 1
i = 2 and j = 2
i = 2 and j = 3
i = 2 and j = 4

-----
Process exited after 0.3118 seconds with
Press any key to continue . . .
```

Lets dry run it with input of 3

```
#include <iostream>
using namespace std;
int main()
{
    int size;
    cout << "\n\n Print a pattern like square with # character:\n";
    cout << "-----\n";
    cout << " Input the number of characters for a side: ";
    cin >> size;
    for (int row = 1; row <= size; ++row)
    {
        for (int col = 1; col <= size; ++col)
        {
            cout << "# ";
        }
        cout << endl;
    }
    return 0;
}
```

With input of 6



D:\Books-ICT\C-Programs\nested_loop2.exe

Print a pattern like square with # character:

Input the number of characters for a side: 6

```
# # # # # #  
# # # # # #  
# # # # # #  
# # # # # #  
# # # # # #  
# # # # # #
```

Process exited after 2.611 seconds with return value 0

Press any key to continue . . .

Changing the inner loop

```
#include <iostream>
using namespace std;
int main()
{
    int size;
    cout << "\n\n Print a pattern like square with # character:\n";
    cout << "-----\n";
    cout << " Input the number of characters for a side: ";
    cin >> size;
    for (int row = 1; row <= size; ++row)
    {
        for (int col = row+1; col <= size; ++col)
        {
            cout << "# ";
        }
        cout << endl;
    }
    return 0;
}
```

With input of 6

 D:\Books-ICT\C-Programs\nested_loop2.exe

Print a pattern like square with # character:

Input the number of characters for a side: 6

```
# # # # #  
# # # #  
# # #  
# #  
#
```

Process exited after 2.496 seconds with return value 0
Press any key to continue . . .

Let's check this out..

```
#include <iostream>
using namespace std;

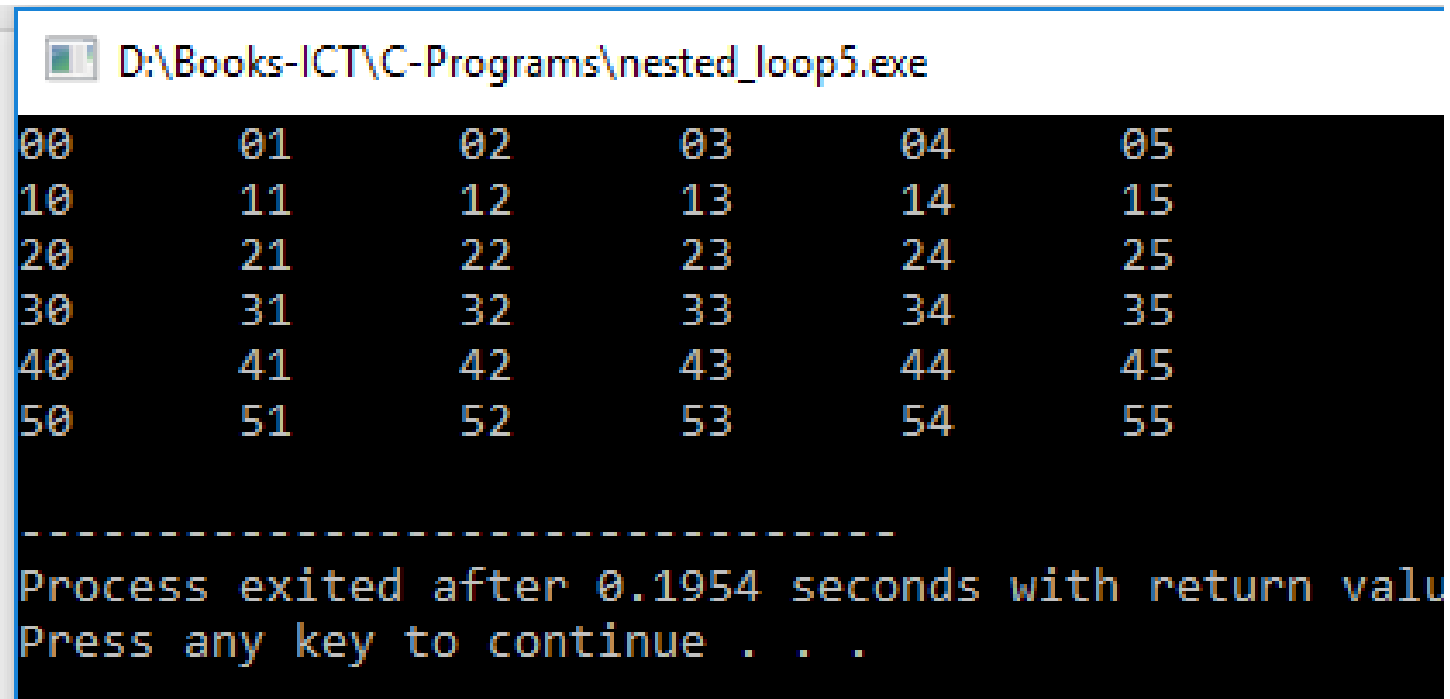
int main ()
{
    int i, j;

    for(i=0; i<=5; i++)
    {

        for(j=0; j <= 5; j++)
            {
                cout << i << j <<" \t";
            }

        cout <<"\n";
    }
    return 0;
}
```

Output..



A screenshot of a Windows command prompt window. The title bar shows the file path "D:\Books-ICT\C-Programs\nested_loop5.exe". The command prompt displays a 6x6 grid of numbers from 00 to 55, followed by a dashed line and a message indicating the process has exited after 0.1954 seconds.

```
D:\Books-ICT\C-Programs\nested_loop5.exe  
00      01      02      03      04      05  
10      11      12      13      14      15  
20      21      22      23      24      25  
30      31      32      33      34      35  
40      41      42      43      44      45  
50      51      52      53      54      55  
  
-----  
Process exited after 0.1954 seconds with return value  
Press any key to continue . . .
```

More practice..

```
#include <iostream>
using namespace std;

int main ()
{
    int i, j;

    for(i=0; i<=5; i++)
    {
        for(j=0; j <= 5; j++)
        {
            cout << i;
        }

        cout << "\n";
    }
    return 0;
}
```

Soln.

 D:\Books-ICT\C-Programs\nested_loop6.exe

000000

111111

222222

333333

444444

555555

Process exited after 0.2223 seconds w

Press any key to continue . . .

Try this..

```
#include <iostream>
using namespace std;

int main()
{
    int n, i, j;

    cout << "Enter number of rows: ";
    cin >> n;

    for(i = 1; i <= n; i++)
    {
        //print * equal to row number
        for(j = 1; j <= i; j++)
        {
            cout << "* ";
        }
        cout << "\n";
    }
    return 0;
}
```

Lets see this..

```
D:\Books-ICT\C-Programs\nested_loop7.exe  
Enter number of rows: 20  
*  
* *  
* * *  
* * * *  
* * * * *  
* * * * * *  
* * * * * * *  
* * * * * * * *  
* * * * * * * * *  
* * * * * * * * * *  
* * * * * * * * * * *  
* * * * * * * * * * * *  
* * * * * * * * * * * * *  
* * * * * * * * * * * * * *  
* * * * * * * * * * * * * * *  
* * * * * * * * * * * * * * * *  
* * * * * * * * * * * * * * * * *  
* * * * * * * * * * * * * * * * * *  
* * * * * * * * * * * * * * * * * * *  
* * * * * * * * * * * * * * * * * * * *  
  
-----  
Process exited after 4.05 seconds with re  
Press any key to continue . . .
```

Book Reference

- D.S. Malik (ch - 4)
- Tony Gaddis (ch - 4)